

Shell Scripting Basics

❖ Task 1: Your First Script

- Create a file hello.sh
- Add the shebang line `#!/bin/bash` at the top
- Print Hello, Devops! Using echo
- Make it executable and run it.

❖ Document: What happens if you remove the shebang line?

- Nothing happened it worked.

```
ubuntu@ip-172-31-47-149:~$ ./hello.sh
Hello, Devops!
ubuntu@ip-172-31-47-149:~$ vim hello.sh
ubuntu@ip-172-31-47-149:~$ ./hello.sh
Hello, Devops!
```

❖ Task 2: Variables

- A variable for your NAME
- A variable for your ROLE
- Print: “Hello, I am \$NAME and I am a \$ROLE”
- Try using single quotes vs double quotes – What is the difference?
 - The single quotes directly print the Hello, I am \$NAME and I am \$ROLE

```
ubuntu@ip-172-31-47-149:~$ vim variables.sh
ubuntu@ip-172-31-47-149:~$ ./variables.sh
Hello, I am Rohan and I am a DevOps Engineer
ubuntu@ip-172-31-47-149:~$ vim variables.sh
ubuntu@ip-172-31-47-149:~$ ./variables.sh
Hello, I am $NAME and I am a $ROLE
ubuntu@ip-172-31-47-149:~$
```

❖ Task 3: User Input with read

- Create greet.sh that:
- Asks the user for their name using read
- Asks for their favorite tool

- Prints: Hello <name>, your favorite tool is <tool>
 - `#!/bin/bash`
 - `# This script is for variables and user input with read`
 - `read -p "Enter your name:" NAME`
 - `read -p "Enter your favourite tool:" TOOL`
 - `echo "Hello $NAME, your favourite tool is $TOOL"`

```
ubuntu@ip-172-31-47-149:~$ chmod 764 greet.sh
ubuntu@ip-172-31-47-149:~$ ./greet.sh
Enter your name:Rohan
Enter your favourite tool:EC2
Hello Rohan, your favourite tool is EC2
```

❖ Task 4: If-Else Conditions

- Takes a number using read
- Prints whether it is positive, negative, or zero
 - `#!/bin/bash`
 - `read -p "Enter the number:" number`
 - `if [ $number -gt 0 ]; then`
 - `echo "The Number is positive."`
 - `elif [ $number -lt 0 ]; then`
 - `echo "The Number is negative."`
 - `else`
 - `echo "The number is zero."`
 - `fi`

```

The number is zero.
ubuntu@ip-172-31-47-149:~$ vim check_number.sh
ubuntu@ip-172-31-47-149:~$ ./check_number.sh
Enter the number:-5
The Number is negative.
ubuntu@ip-172-31-47-149:~$ ./check_number.sh
Enter the number:12
The Number is positive.

```

- Create file_check.sh that:
- Ask for a filename
- Check if the file exists using -f
- Prints appropriate message
 - `#!/bin/bash`
 - `read -p "Enter the file path to be found" file_path`
 -
 - `if [ -f $file_path ]; then`
 - `echo "File Exists"`
 - `else`
 - `echo "File Doesn't Exists"`
 - `fi`

```

ubuntu@ip-172-31-47-149:~$ ./file_check.sh
Enter the file path to be found/etc/passwd
File Exists
ubuntu@ip-172-31-47-149:~$ ./file_check.sh
Enter the file path to be foundhello.sh
File Exists
ubuntu@ip-172-31-47-149:~$ ./file_check.sh
Enter the file path to be foundgreet.sh
File Exists

```

❖ Task 5: Combine It all

- Create server_check.sh that:

- Stores a service name in variable (eg, nginx, sshd)
- Ask the user: “Do you want to check the status? (y/n)”
- If y – runs `systemctl status <service>` and prints whether it is active or not
- If n – prints “skipped”
 - `#!/bin/bash`
 - `SERVICE="sshd"`
 - `read -p "Do you want to check the status of$SERVICE? (y/n):"`
`choice`
 - `if [ "$choice" == "y" ]; then`
 - `systemctl is-active --quiet $SERVICE`
 - `if [ $? -eq 0 ]; then`
 - `echo "$SERVICE is active"`
 - `else`
 - `echo "$SERVICE is not active"`
 - `fi`
 - `else`
 - `echo "Skipped."`
 - `fi`

```
ubuntu@ip-172-31-47-149:~$ vim server_check.sh
ubuntu@ip-172-31-47-149:~$ ./server_check.sh
Do you want to check the status of sshd? (y/n): y
sshd is not active
ubuntu@ip-172-31-47-149:~$ ./server_check.sh
Do you want to check the status of sshd? (y/n): n
skipped.
```

❖ What I learned (3 key points)

- If we are printing only lines, there is no need to add shebang; it will work.
- Hashtags are used to comment the lines.
- We can create file_check.sh shell script to check whether file exists or not.